AI Developer Technical Challenge

Overview

Build a Python application with a command-line agent that books a restaurant table through conversation.

The restaurants are a small JSON file we give you. Restaurant data must be obtained through the functions, not placed in the prompt.

Expected time: around 2 hours.

The functions

We provide `tools.py` and `restaurants.json`. Use them as is — they are not part of the evaluation.

- `search_restaurants(cuisine, area, party_size, max_price)` — all arguments optional
- `get_details(restaurant_id)` — includes tonight's free times
- `book_table(restaurant_id, time, party_size)` — returns a booking reference

The agent

The agent decides by itself which functions to call and when, and keeps talking to the user until there is a booking, or a clear answer that there is none.

It must handle a conversation like this:

1. "Dinner tonight, somewhere Italian."
2. "Four of us."
3. "What time can we get in?"
4. "Book that one."

And it must:

- ask for what it needs in order to search, one question at a time;
- remember what it already offered, so that "that one" means something;
- confirm the restaurant, the time and the number of people before booking;

- only state names, times and prices that came back from the functions;
- when a booking is not possible, say why, rather than offering something it has not checked.
- handle the table being taken between offering it and booking it;

Deliverables

Please include:

- the code;
- a README describing:
    - how to run it;
    - any assumptions you made;
    - Model chosen;

If you encounter ambiguous or unspecified requirements, make the assumptions you consider appropriate and document them in the README.

Use any LLM provider you have access to.

Complete as much as possible in the given expected time.